

# D.A.R.I.A. – Domain-Specific Agentic Research, Intelligence & Analysis System

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*Memory-Driven Hybrid Research, Reflection and Summarization using LangGraph, RAG, Web Search and Persistent Memory*

## Abstract

D.A.R.I.A. is a domain-specific agentic research system designed to address limitations of traditional Retrieval-Augmented Generation (RAG) systems. The platform integrates persistent memory, research planning, hybrid retrieval, evidence curation, reflection loops, and structured summarization into a unified LangGraph workflow. Rather than producing immediate responses from retrieved information, the system follows a multi-stage research process that resembles how a human researcher plans, gathers evidence, evaluates quality, revises strategy, and synthesizes findings. The result is an explainable, auditable, and continuously improving research workflow.

## 1. Introduction

Recent advances in Large Language Models have enabled powerful question-answering systems. However, most systems remain fundamentally reactive. Traditional RAG pipelines retrieve information and generate responses but lack long-term memory, self-evaluation, and adaptive research strategies. D.A.R.I.A. was developed to bridge this gap by introducing memory-driven planning, critique-based reflection, and persistent learning.

The project demonstrates how agentic AI principles can be applied to research workflows. By combining multiple specialized agents with state management and reflection mechanisms, the system transforms research from a one-shot retrieval process into an iterative and evidence-driven workflow.

## 2. Problem Statement

Conventional RAG systems face several challenges:

- No memory of previous research sessions.
- Limited transparency regarding reasoning and retrieval decisions.
- No mechanism for evaluating evidence quality.
- Inability to revise search strategies after failure.
- Lack of structured research planning.

D.A.R.I.A. addresses these challenges through persistent memory, planning agents, evidence curation, critic-driven evaluation, reflection loops, and structured reporting.

### 3. System Architecture

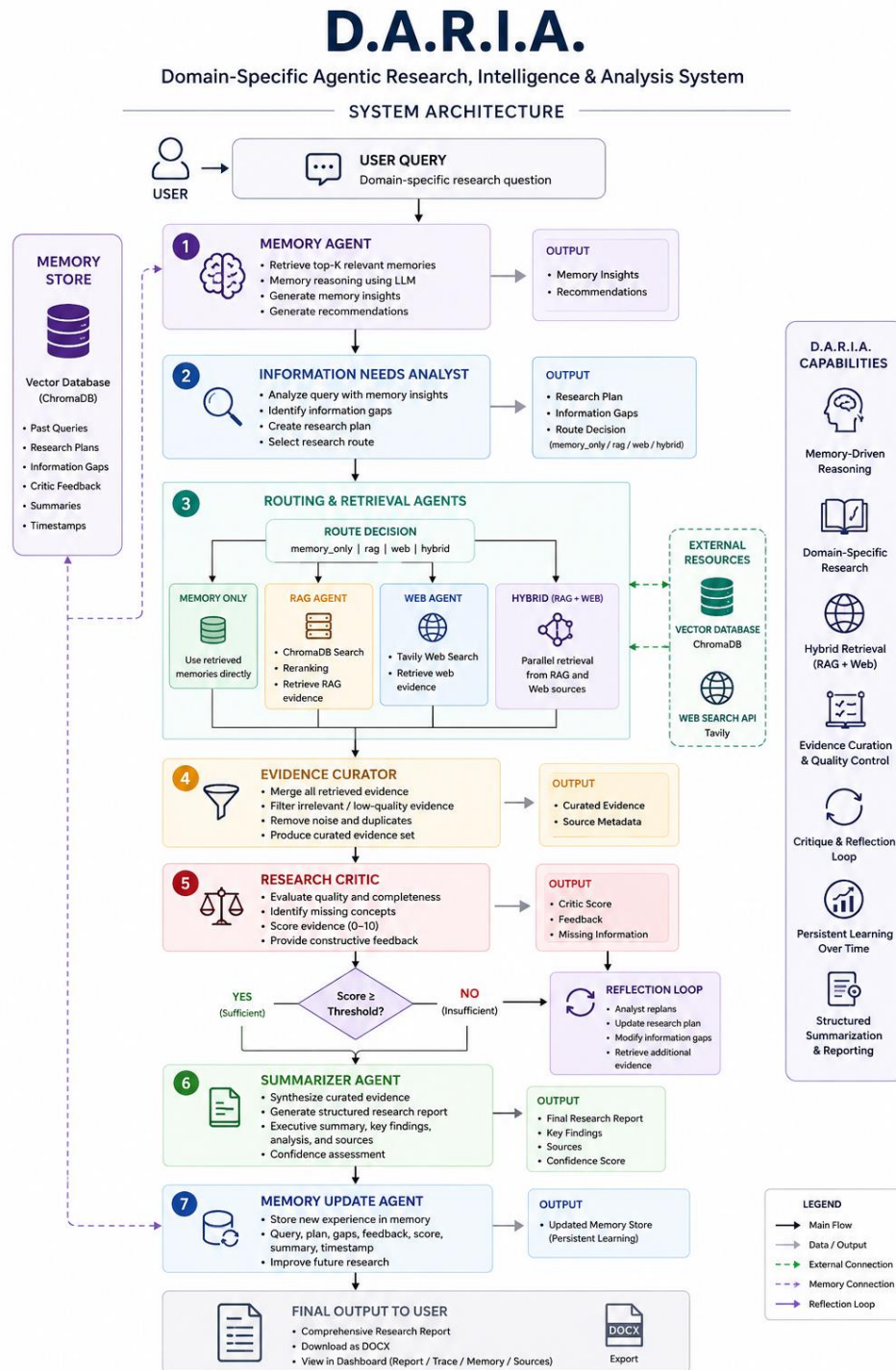


Figure 1. D.A.R.I.A. System Architecture

The architecture consists of five major layers:

**Memory Layer:**

Stores previous research experiences including plans, information gaps, critic feedback, scores, and summaries.

**Planning Layer:**

Analyzes user intent, identifies information gaps, and selects an optimal research route.

**Retrieval Layer:**

Performs domain-specific RAG retrieval, web search, or hybrid retrieval through parallel execution.

**Evaluation Layer:**

Filters evidence and evaluates research quality using a dedicated critic agent.

**Learning Layer:**

Stores research outcomes and feedback to improve future investigations.

## **4. LangGraph Workflow Design**

D.A.R.I.A. is implemented using LangGraph and a shared ResearchState object. Each node is responsible for a specialized task while state transitions are managed through conditional routing.

**Workflow Characteristics:**

- Conditional routing.
- Parallel execution.
- Reflection loops.
- State persistence.
- Memory integration.

**Routing Modes:**

1. memory\_only
2. rag
3. web
4. hybrid

The hybrid route executes RAG and Web Agents concurrently and merges results before evaluation.

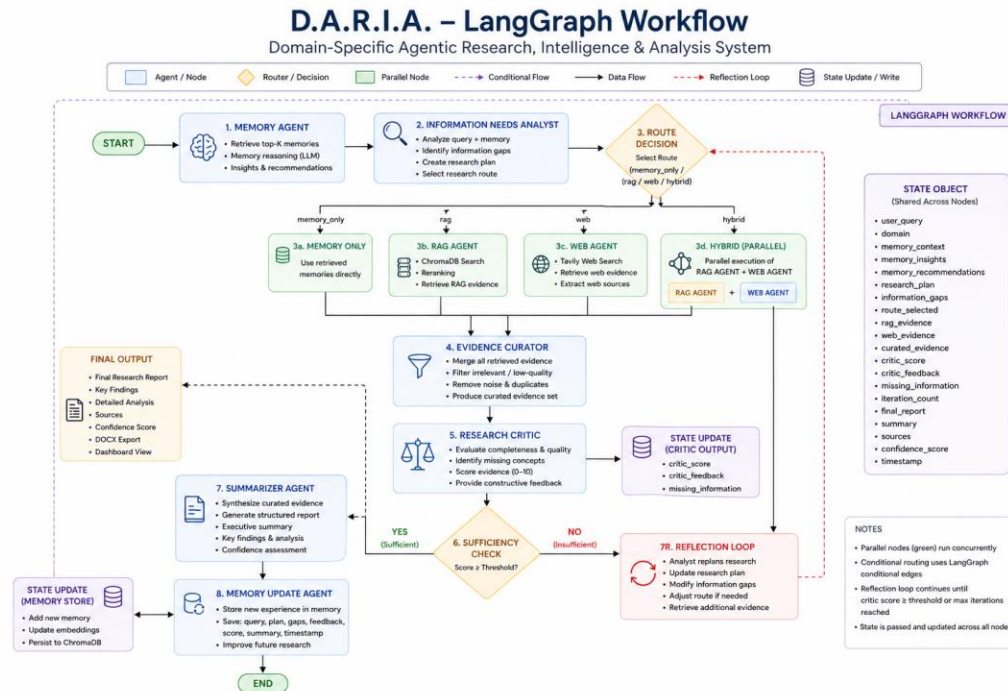


Figure 2. LangGraph Workflow and Conditional Routing

## 5. Agent Design

### 5.1 Memory Agent

The Memory Agent retrieves previous research sessions and generates memory insights and recommendations. Retrieved memories influence planning and help avoid repeating failed research strategies.

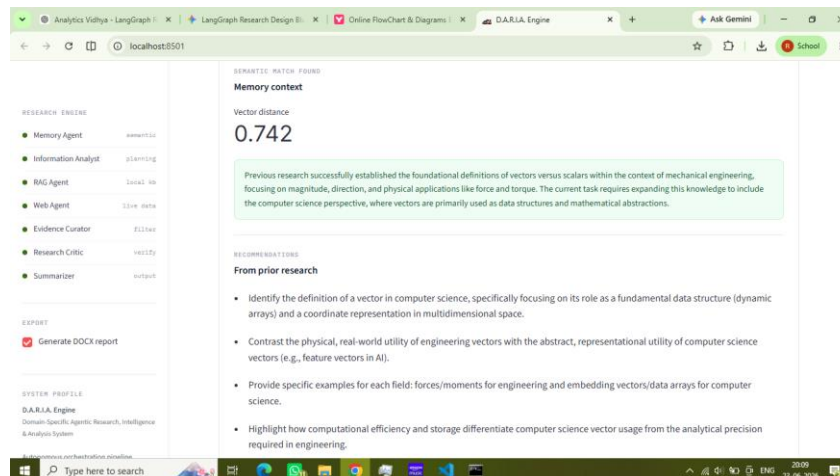


Figure 3: Memory Summary

## 5.2 Information Needs Analyst

The Information Needs Analyst converts user questions into structured research plans. It identifies missing concepts, determines information gaps, and selects retrieval strategies.

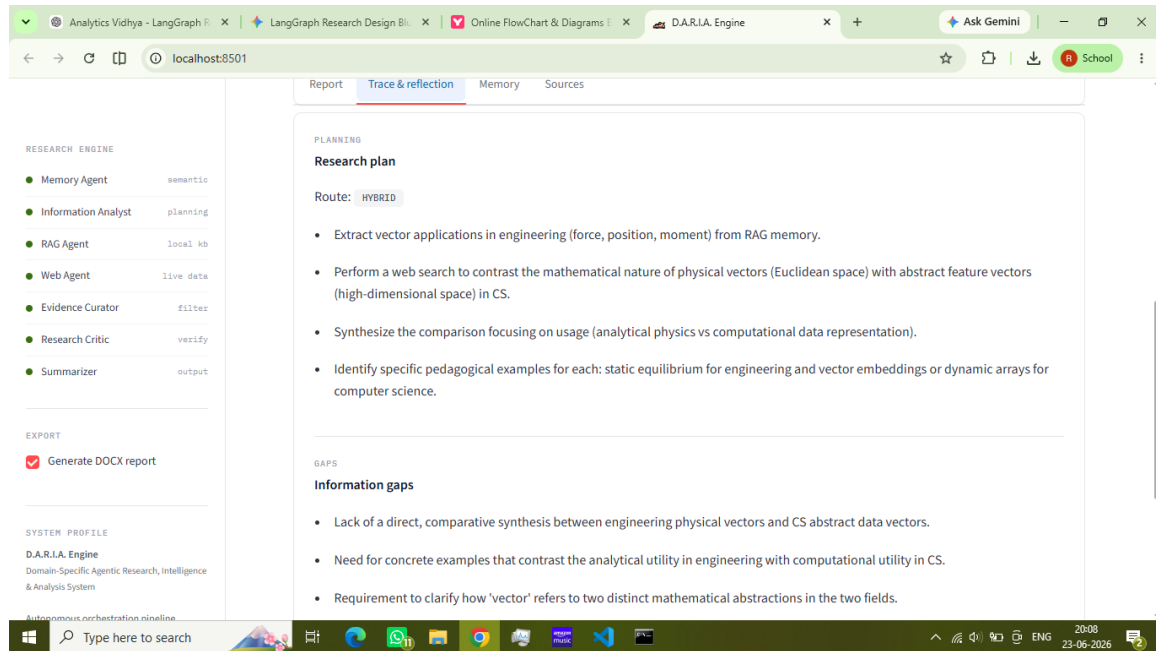


Figure 4: Research Planning

## 5.3 RAG Agent

The RAG Agent retrieves domain-specific knowledge from a ChromaDB vector store built from engineering mechanics resources. Semantic search and reranking improve evidence quality.

## 5.4 Web Agent

The Web Agent retrieves dynamic and recent information using Tavily search. It complements domain-specific retrieval when knowledge is unavailable within the corpus.

## 5.5 Hybrid Retrieval

Hybrid retrieval executes RAG and Web Agents concurrently. This improves coverage while maintaining domain relevance.

```
Select Command Prompt - streamlit run streamlit_app.py
ical representational tool.
- Structure future comparisons to contrast physical application (force/torque) with computational application (pattern recognition/spatial rendering).
Route: hybrid
Search Query: vector applications mechanical engineering vs computer science machine learning embeddings computer graphics linear algebra differences

Information Gaps:
- Specific examples of linear algebra applications in computer graphics (e.g., coordinate transformation matrices).
- Conceptual difference between vectors as physical entities and vectors as abstract data structures or high-dimensional feature representations.
- Computational complexity implications of vector usage in engineering simulation versus data science feature processing.

Research Plan:
- Synthesize RAG knowledge on vectors as physical quantities (force, torque, displacement) in statics and dynamics.
- Search for computer science applications: linear algebra in computer graphics (transform matrices) and embeddings in ML/Data Science.
- Identify computational complexity distinctions: vectors as physical entities in simulation vs. vectors as data structures/high-dimensional arrays in algorithms.
- Formulate a structured comparison contrasting physical force/torque application with computational spatial rendering and pattern recognition.

[Routing] Hybrid research using parallel RAG + Web...

[RAG Agent] Retrieving knowledge...
[RAG Agent] Query:
vector applications mechanical engineering vs computer science machine learning embeddings computer graphics linear algebra differences

Specific examples of linear algebra applications in computer graphics (e.g., coordinate transformation matrices). Conceptual difference between vectors as physical entities and vectors as abstract data structures or high-dimensional feature representations. Computational complexity implications of vector usage in engineering simulation versus data science feature processing.

[Web Agent] Searching web...
[Web Agent] Query:
vector applications mechanical engineering vs computer science machine learning embeddings computer graphics linear algebra differences
[Web Agent] Query: vector applications mechanical engineering vs computer science machine learning embeddings computer graphics linear algebra differences
[Web Agent] Results Found: 3

[Evidence Curator] Filtering evidence...
[Evidence Curator] Total Evidence: 6Kept: 6 | Discarded: 0
[Evidence Curator] Reasoning: These evidence items provide definitions, explanations, and examples of vectors in both mechanical engineering and computer science, and demonstrate how vectors are used in different fields. They include textbook excerpts that define vectors and explain how to resolve them into scalar components, as well as web pages that discuss the applications of vectors in computer science, such as computer graphics, machine learning, and quantum computing.

[Research Critic] Evaluating Filtered evidence...
Evidence Count: 7
Compiled Evidence Context:
```

Figure 5: Hybrid Retrieval

## 5.6 Evidence Curator

The Evidence Curator merges and filters evidence before evaluation. Irrelevant, redundant, or weak evidence is discarded.

```
Select Command Prompt - streamlit run streamlit_app.py
- Synthesize the comparison focusing on usage (analytical physics vs computational data representation).
- Identify specific pedagogical examples for each: static equilibrium for engineering and vector embeddings or dynamic arrays for computer science.

[Routing] Hybrid research using parallel RAG + Web...

[RAG Agent] Retrieving knowledge...

[Web Agent] Searching web...
[Web Agent] Query:
comparison of physical vectors in mechanical engineering vs abstract data vectors in computer science, examples of force vectors vs feature embedding vectors
[Web Agent] Query: comparison of physical vectors in mechanical engineering vs abstract data vectors in computer science, examples of force vectors vs feature embedding vectors
[Web Agent] Results Found: 3

[Evidence Curator] Filtering evidence...
[Evidence Curator] Total Evidence: 6Kept: 4 | Discarded: 2
[Evidence Curator] Reasoning: These evidence contain definitions, explanations, and examples of vectors in Mechanical engineering and computer science, which are relevant to the question.

[Research Critic] Evaluating Filtered evidence...
Evidence Count: 5
Compiled Evidence Context:

Evidence 2
Source Type:
Textbook
Chapter:
CHAPTER 1: FUNDAMENTAL CONCEPTS
Section:
1.1 Preparatory Concepts
Page:
20
```

Figure 6: Evidence Curation

## 5.7 Research Critic

The Research Critic evaluates completeness, relevance, and evidence quality. It assigns scores and identifies deficiencies.

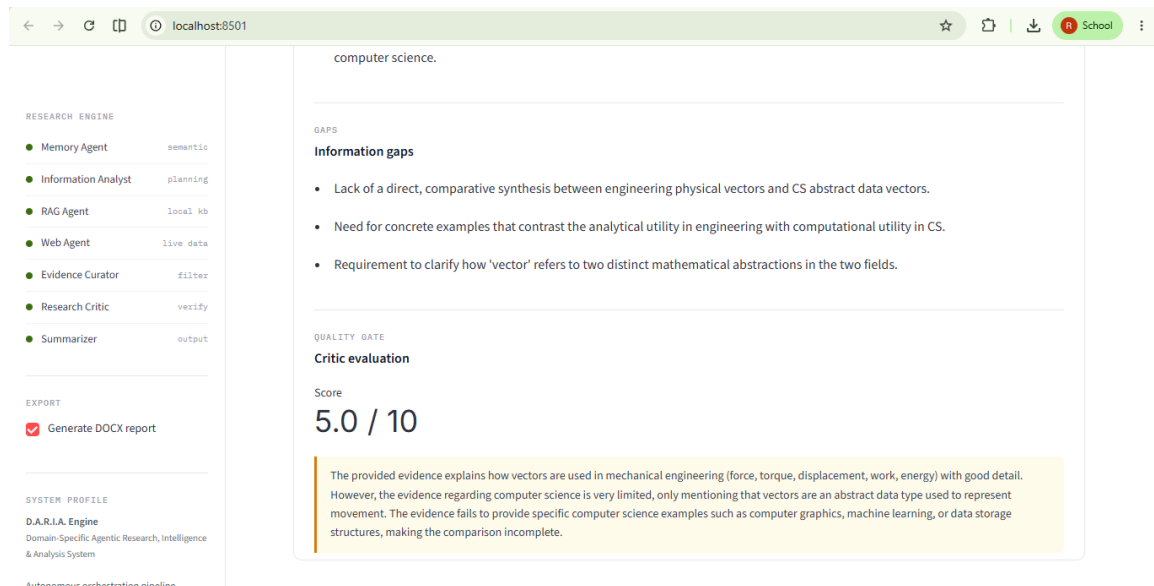


Figure 7: Research Critic

## 5.8 Summarizer Agent

The Summarizer synthesizes validated evidence into structured reports including executive summaries, findings, analysis, sources, and confidence assessments.

## 5.9 Memory Update Agent

The Memory Update Agent stores research experiences, enabling persistent learning across sessions.

## 6. Reflection-Based Research Improvement

A major innovation of D.A.R.I.A. is the reflection loop. After retrieval and curation, evidence is evaluated by the Research Critic. If evidence quality is insufficient, critic feedback is transformed into new information gaps and research plans.

This mechanism enables:

- Self-correction.



- Iterative improvement.
- Adaptive retrieval.
- Increased research completeness.

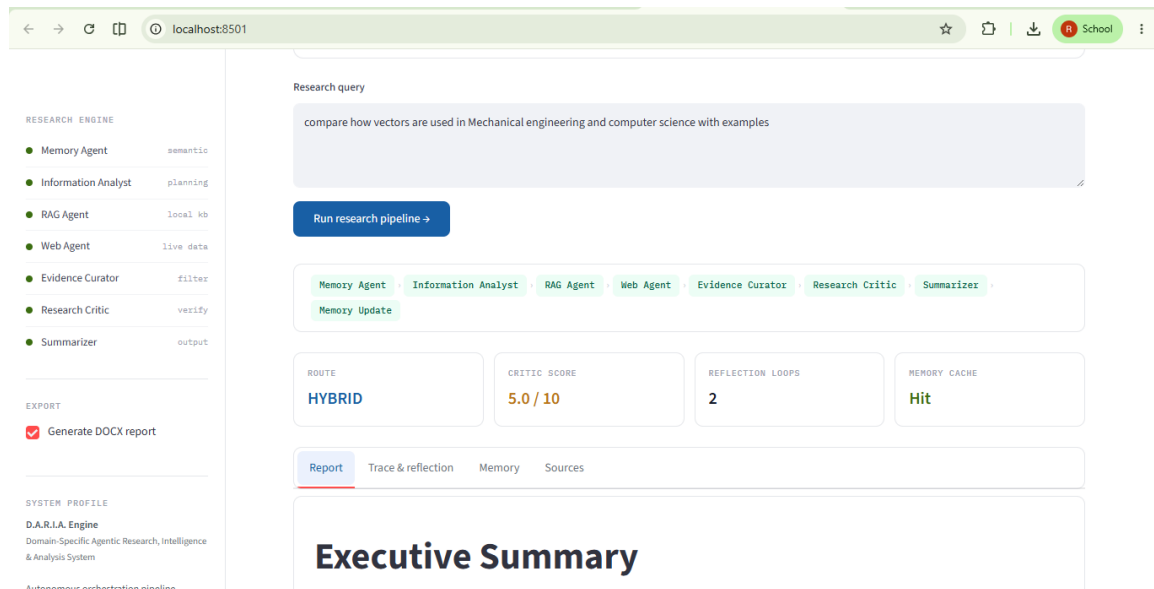


Figure 8. Reflection Loop

## 7. Report Generation and User Interface

The Streamlit interface exposes the entire research workflow. Users can inspect research plans, information gaps, critic feedback, memory insights, and sources.

The final report includes:

- Executive Summary
- Key Findings
- Detailed Analysis
- Sources
- Confidence Assessment

Reports can be exported as DOCX documents.



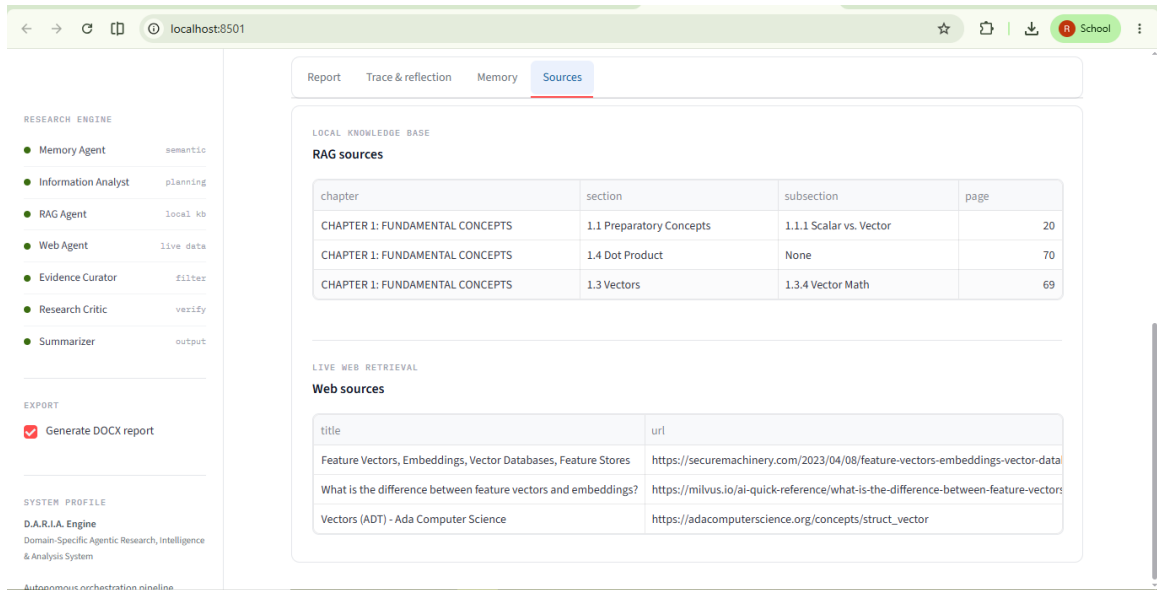


Figure 9: Sources tab

## 8. Technical Stack

Agent Framework: LangGraph

LLM Orchestration: LiteLLM

Research Model: Gemini 2.5 Flash

Vector Database: ChromaDB

Embeddings: BAAI BGE Small

Web Search: Tavily

Frontend: Streamlit

Document Export: python-docx

Knowledge Extraction: Docling

## 9. Key Innovations

### 9.1 Memory-Augmented Research

Traditional RAG systems operate without long-term memory. D.A.R.I.A. stores research plans, information gaps, critic feedback, and summaries. Future sessions benefit from accumulated experience.

### 9.2 Research Planning Layer

The Information Needs Analyst explicitly plans research before retrieval. This shifts the workflow from reactive retrieval to goal-driven investigation.

### 9.3 Corpus-Aware Routing

Different research problems require different retrieval strategies. D.A.R.I.A. dynamically selects memory, RAG, web, or hybrid routes.

### 9.4 Parallel Hybrid Retrieval

Concurrent execution of RAG and Web Agents improves coverage while reducing latency.

### 9.5 Evidence Curation

Evidence is filtered before summarization. This reduces hallucinations and improves answer quality.

### 9.6 Critique-Based Evaluation

The Research Critic acts as an independent evaluator that measures evidence quality and identifies deficiencies.

### 9.7 Reflection Loop

Rather than accepting imperfect evidence, the system replans and retries retrieval. This is one of the most agentic aspects of the architecture.

### 9.8 Persistent Learning

Research outcomes are stored for future use. The system becomes increasingly useful as additional research sessions accumulate.

### 9.9 Explainable Research

Research plans, gaps, scores, and feedback are exposed to users. This creates transparency and trust.

### 9.10 Structured Research Reporting

Instead of generating conversational responses, the system produces professional research reports with traceable sources and confidence assessments.

## 10. Results and Demonstration

Testing demonstrated successful operation of memory retrieval, hybrid retrieval, evidence curation, critic-driven reflection, structured summarization, and DOCX export.

Key outcomes:

- Successful memory integration.
- Dynamic routing.
- Parallel retrieval.
- Reflection-driven improvements.
- Explainable workflows.
- Professional report generation.

## 11. Limitations

Current limitations include:

- Single-domain corpus.
- Threshold-based memory retrieval.
- Fixed reflection iterations.
- Text-centric evidence processing.
- Limited source reliability assessment.

## 12. Future Work (D.A.R.I.A. V2)

Future enhancements include:

- Multi-domain corpora.
- Multimodal RAG.
- Memory relevance ranking.
- Adaptive retrieval strategies.
- Source credibility scoring.
- Autonomous task decomposition.
- Performance analytics dashboards.
- Agent benchmarking.

## 13. Conclusion

D.A.R.I.A. demonstrates how memory, planning, retrieval, critique, reflection, and summarization can be integrated into a unified research workflow. By combining LangGraph orchestration, domain-specific RAG, web search, evidence curation, and persistent memory, the system moves beyond traditional question-answering architectures toward explainable and continuously improving research systems. The project highlights

the potential of agentic AI for engineering education, research support, and knowledge-intensive decision-making.